AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 2 · WORKBOOK

Input, Process, Output

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected information record.

Trace a request, compare summaries, choose needed information, diagnose an error and attempt the reserved new case.

Keep one habit: Keep a first attempt, its supporting trace and a correction you can explain.

MAIN LESSON · SECTION 1

Recall · Follow last week's decision

> Retrieve before checking your notes.

Your first explanation

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

▼ **Earlier wording, alternative practice and picture transcripts**

AI or Not?

MAIN LESSON · SECTION 12

W1 · Trace the library request

> Check both conditions and identify the surviving record.

Read, trace and explain

- The kiosk receives genre=science and maximum pages=120. The catalogue contains A science 80 pages, B science 150 pages and C mystery 90 pages. It keeps records with the requested genre and no more than 120 pages.

  Draw the **input**, **process** and **output**, naming the surviving record.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Two schedules, one total

> Show why the summary cannot determine the busiest hour.

Try the visitor case

- A record stores only total visitors=3. Give two different arrival schedules that produce this total. Can the total tell us the busiest hour? Explain.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Choose enough information

> Match each question to the information needed to answer it.

Explain each choice

- For each task, choose the information needed: decide whether a book is under 120 pages; quote its first sentence exactly; decide whether a copy is available now. Explain why a genre tag alone is insufficient for all three.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Find the first incorrect stage

> Trace the spoken request, transcript and arithmetic before choosing a repair.

Diagnose the voice calculator

- A voice calculator receives a spoken request for 16 divided by 4. Its transcript says 60 divided by 4 and its calculator returns 15.

  Identify the first incorrect stage, show the intended result and propose a check before calculation.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Explain a new information loss

> Keep one first attempt on the four-score case before feedback.

Read and explain independently

A classroom device receives four scores and stores only their total, 20. A student claims that every score must be 5. Give two different lists of four nonnegative whole-number scores with total 20.

Explain precisely what the stored total establishes and what it does not.

> **My independent explanation, trace and limits**

Record any support

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternative practice and picture transcripts**

Independent case: Investigate a new loss of information

Project · Draw your sorter’s information path

> Label the input, transformation, output and human review point.

Your map and evidence

- Draw the Classroom Sorter's first **input-process-output** map and show where a person may review the result.
- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair, then return

> Keep the first attempt, the reason for a repair and a later changed-case check.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Check what your work shows

> Point to a trace, an explanation and a limit before choosing a confidence level.

Your self-check

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Visitor practice record**

REFERENCE · SECTION 2

Reference · Visitor practice record

> This is the same W2 attempt. Use one response record.

Use this when helpful

Use W2: Finish the visitor example for **Representation**s and information paths. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Is every output final?

REFERENCE · SECTION 3

Extra practice · Is every output final?

> Use a multi-stage example to repair the claim.

Use this when helpful

Use the worked case for **Representation**s and information paths. Proposed explanation: "Every **output** is the final action". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

▼ Reference · The kiosk target

REFERENCE · SECTION 5

Reference · The kiosk target

> Return to the target when checking your explanation.

Use this when helpful

A library kiosk receives 'mystery', filters a catalogue by genre and displays three titles. No speaker is attached. Draw its information path. A visitor says it read three books aloud. Which stage contradicts that claim?

▼ **Extra practice · Playlist helper**

REFERENCE · SECTION 6

Extra practice · Playlist helper

> Try before checking the student worked explanation.

Use this when helpful

A playlist helper receives a mood, matches song tags and displays three titles.

Name **input**, **process** and **output**. Is displaying titles the same as playing audio?

Show your **trace**, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · Fan and missing reading**

REFERENCE · SECTION 7

Extra practice · Fan and missing reading

> This is additional practice; keep the four-score first attempt separate.

Use this when helpful

A classroom fan receives a temperature reading, compares it with 25 degrees C, and turns on. Name the **input**, **process** and **output**. What should happen if the reading is missing?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 2 Test and reflect

▼ **Extra practice · Design a useful test**

REFERENCE · SECTION 8

Extra practice · Design a useful test

> Write a prediction before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Kiosk response space**

REFERENCE · SECTION 11

Reference · Kiosk response space

> The kiosk was used in teaching. This response is practice.

Use this when helpful

A library kiosk receives 'mystery', filters a catalogue by genre and displays three titles. No speaker is attached. Draw its information path. A visitor says it read three books aloud. Which stage contradicts that claim?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

▼ Reference · Four-score response copy

REFERENCE · SECTION 15

Reference · Four-score response copy

> This duplicates W4. Reuse its first-attempt record.

Use this when helpful

A classroom device receives four scores and stores only their total, 20. A student claims that every score must be 5. Give two different lists of four nonnegative whole-number scores with total 20.

Explain precisely what the stored total establishes and what it does not.

> **My reasoning and evidence**

▼ Extra practice · The pencil record

REFERENCE · SECTION 18

Extra practice · The pencil record

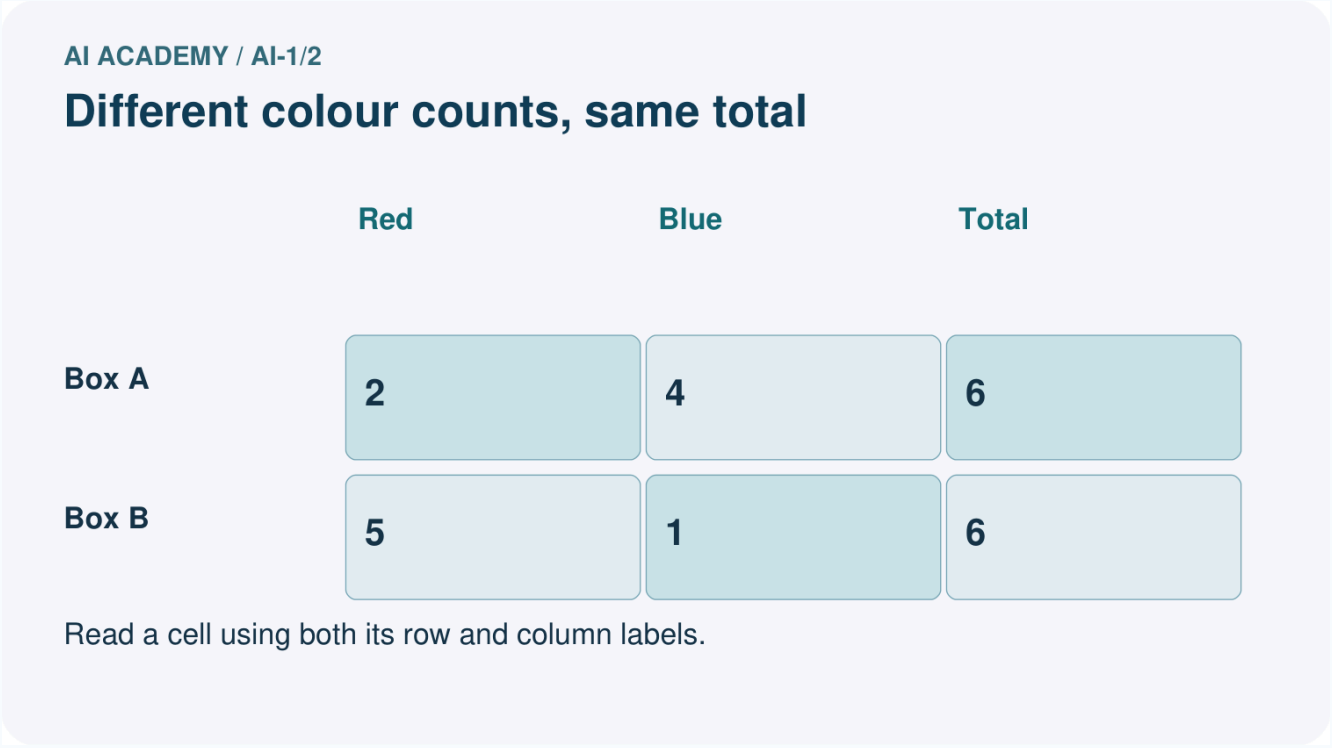
> Use the worked picture and record what support you needed.

Use this when helpful

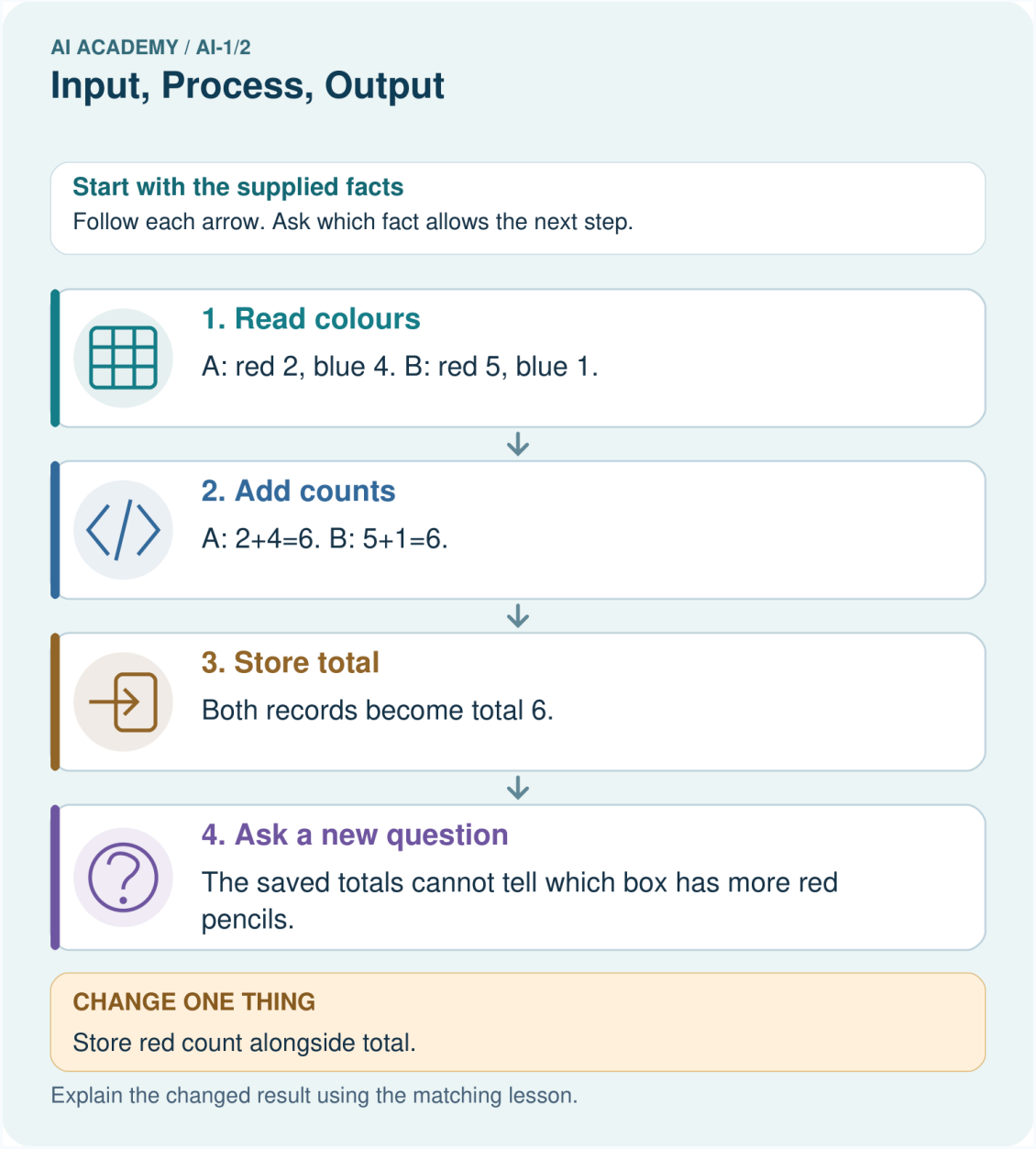
A fictional classroom cupboard records coloured pencils, then reports just a total.

Box A has 2 red and 4 blue pencils. Box B has 5 red and 1 blue.

The **process** adds colour counts and stores only the total, not the two counts.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Choose sorter **input**s that retain the clues needed for its decision; draw the transformation and identify discarded details.

▼ **Reference · Sorter portfolio record**

REFERENCE · SECTION 19

Reference · Sorter portfolio record

> Extend your main project record if needed.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Choose sorter **input**s that retain the clues needed for its decision; draw the transformation and identify discarded details.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one paper or browser copy.

First supported question

- A fictional classroom cupboard records coloured pencils, then reports just a total.
- Which labelled step matches this detail? The saved totals cannot tell which box has more red pencils.
- 1. Add counts
- 2. Read colours
- 3. Ask a new question
- 4. Store total
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Ask a new question The saved totals cannot tell which box has more red pencils.

Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct **input**s lead to the same stored **output**.

Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally.

Changed-record question

- Changed-case practice: Store red count alongside total. Which conclusion is supported by the supplied facts?
- 1. The records become (6,2) and (6,5).
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed record**

After your attempt, compare with the supported explanation: The records become (6,2) and (6,5). Blue count can now be recovered by total minus red, assuming these are the only colours.

Repair and save your work

Review AI or Not? (ai-1/1). Goal: Distinguish an observed result from an explanation of how it was produced. Short practice: Point to the learned stage. Does the changed display prove the weather forecast changed?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternative practice and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

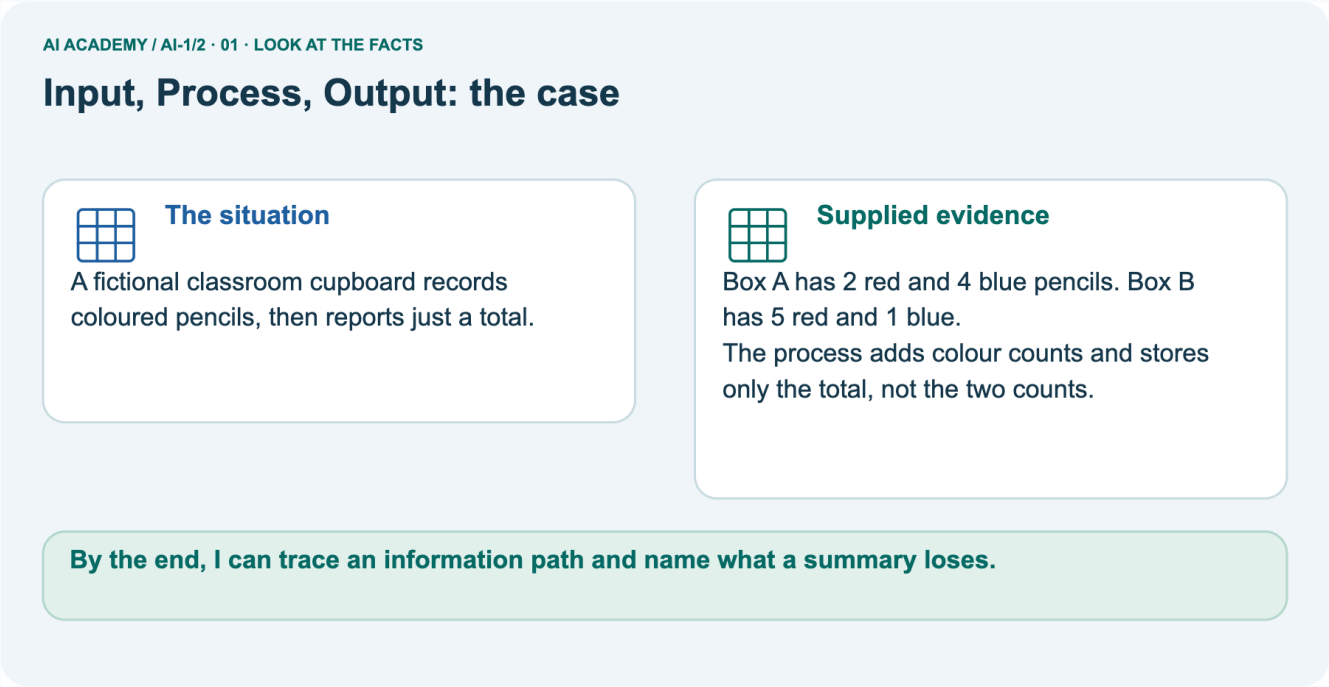
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture rather than completing all six again.

01. Input, Process, Output: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

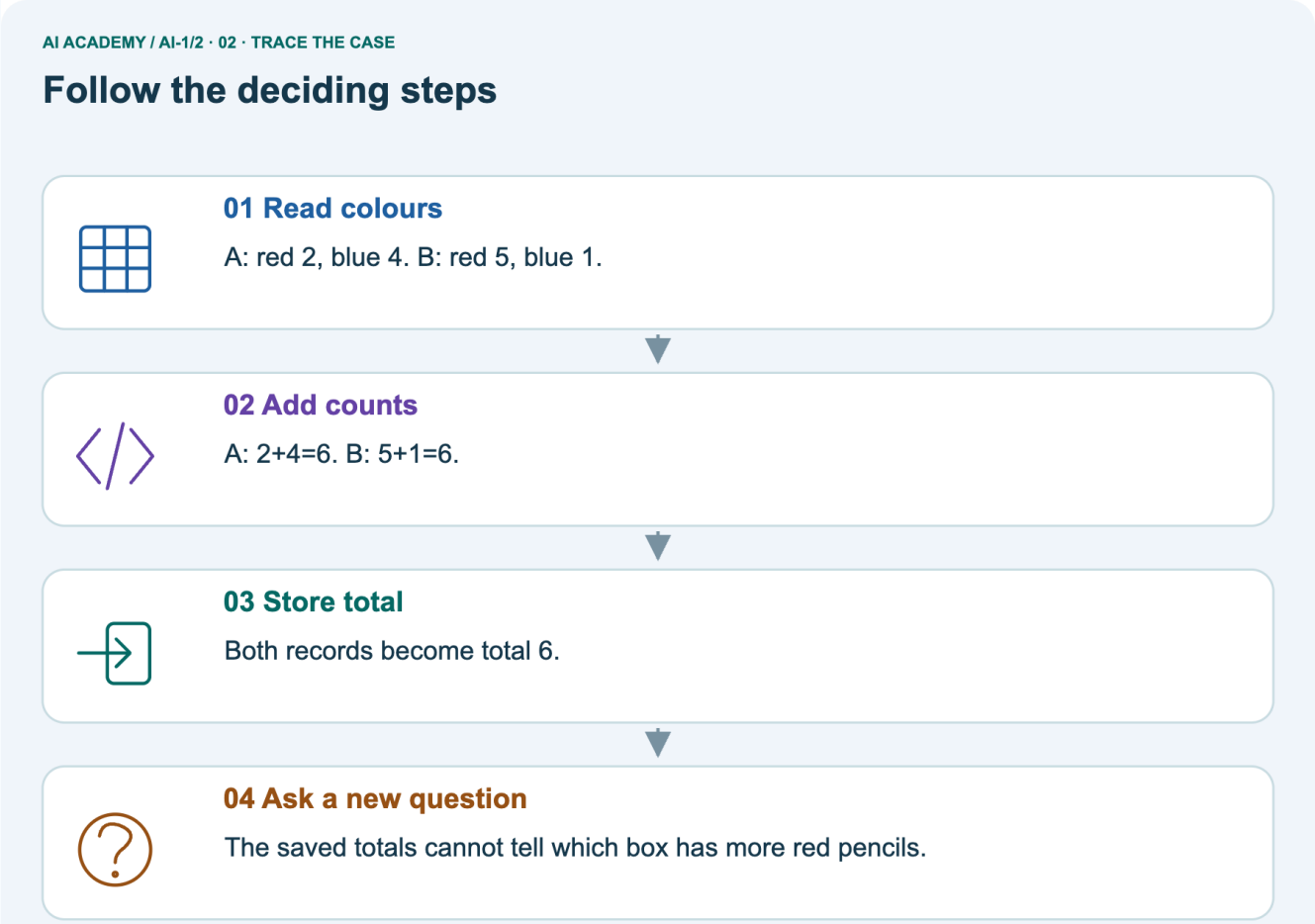
A fictional classroom cupboard records coloured pencils, then reports just a total. Supplied facts: Box A has 2 red and 4 blue pencils. Box B has 5 red and 1 blue.

The **process** adds colour counts and stores only the total, not the two counts.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



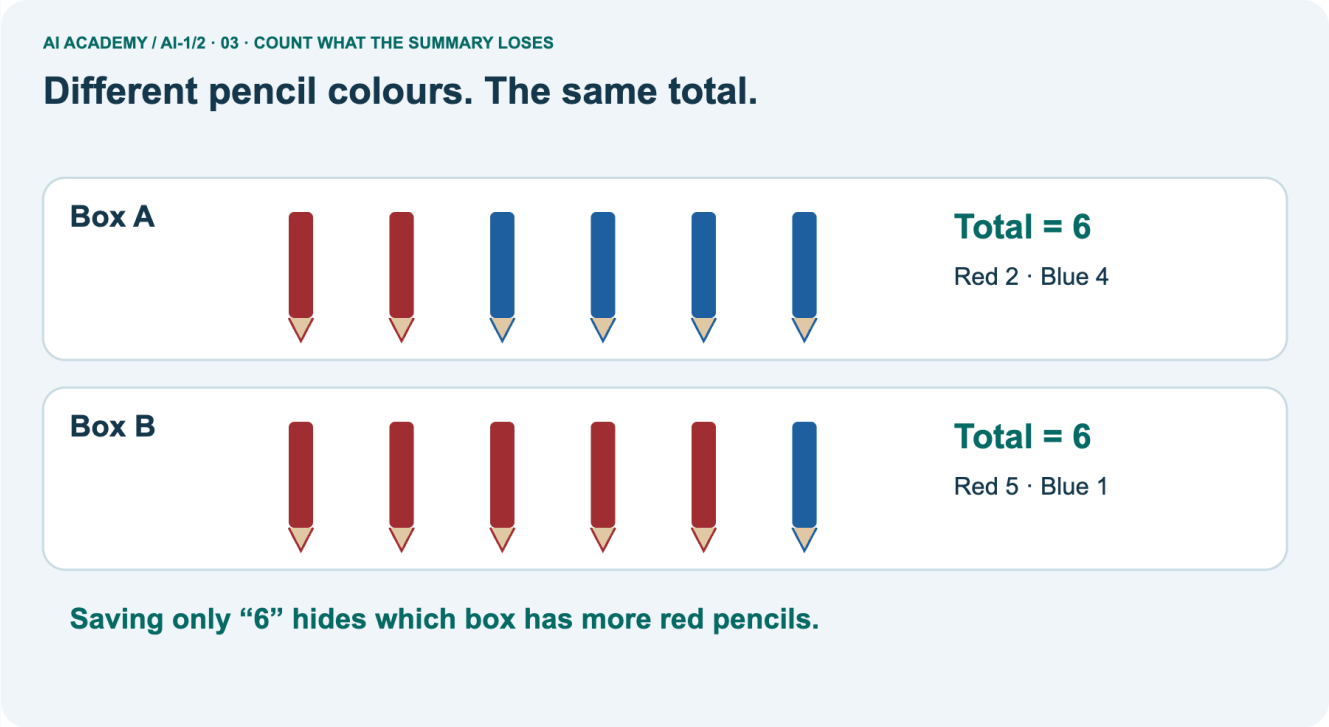
Follow the labelled picture with the matching explanation. Open the vector diagram

Read colours: A: red 2, blue 4. B: red 5, blue 1. → Add counts: A: 2+4=6. B: 5+1=6. → Store total: Both records become total 6. → Ask a new question: The saved totals cannot tell which box has more red pencils.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Different pencil colours. The same total.



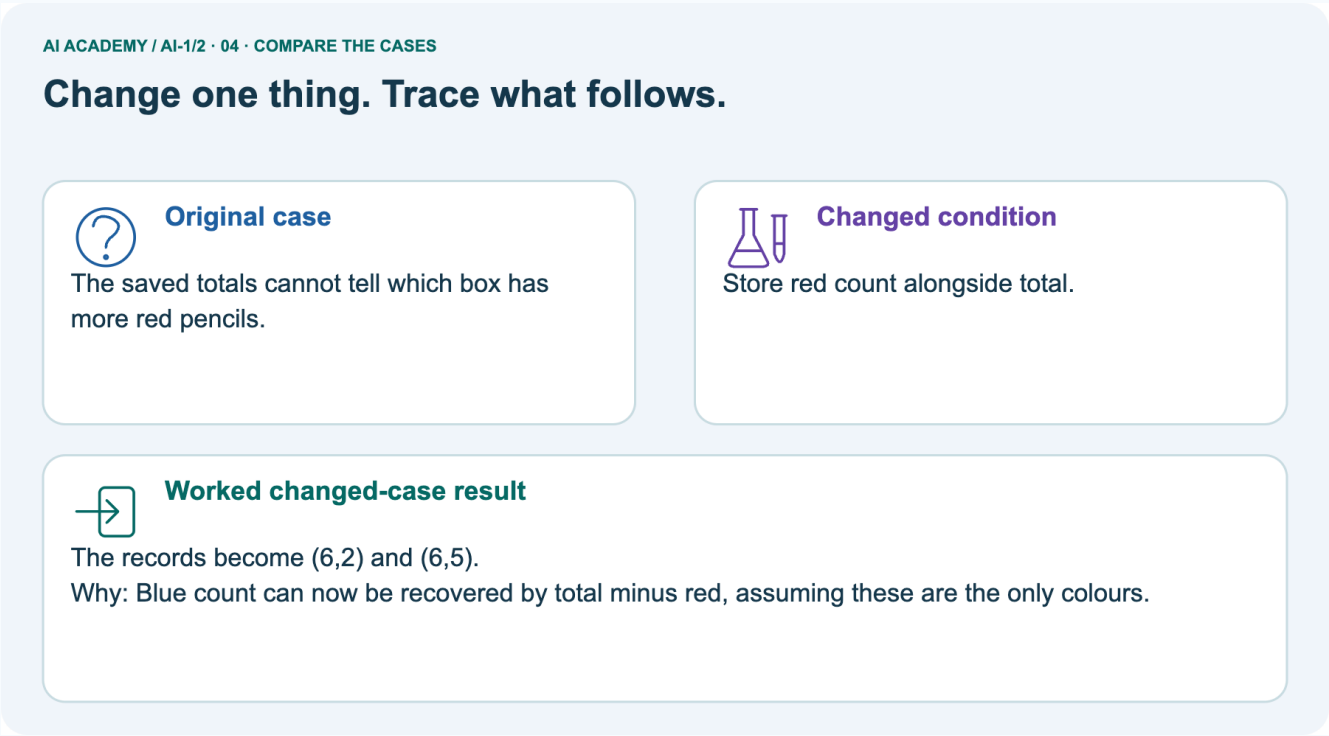
Follow the labelled picture with the matching explanation. Open the vector diagram

Box A: 2 red pencils and 4 blue, total 6. Box B: 5 red pencils and 1 blue, total 6. The equal totals do not preserve the colour counts.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

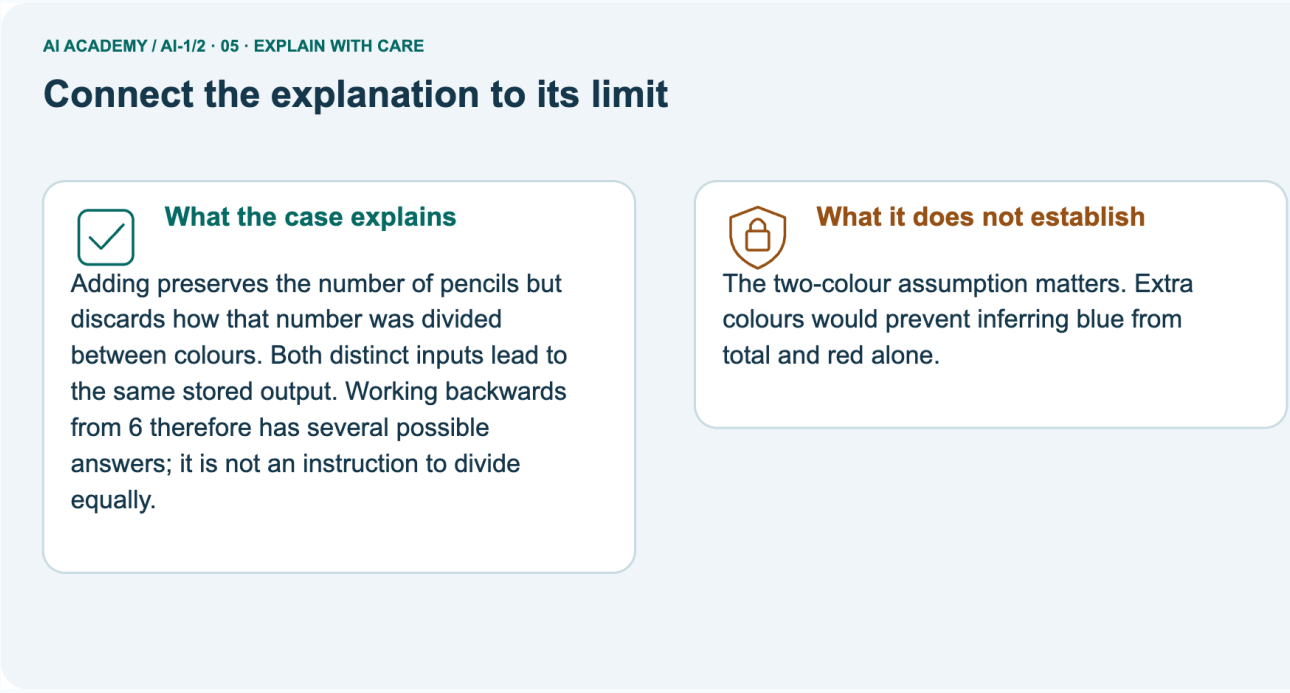
Original case: The saved totals cannot tell which box has more red pencils. Changed condition: Store red count alongside total. Expected result: The records become (6,2) and (6,5).

Reason: Blue count can now be recovered by total minus red, assuming these are the only colours.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

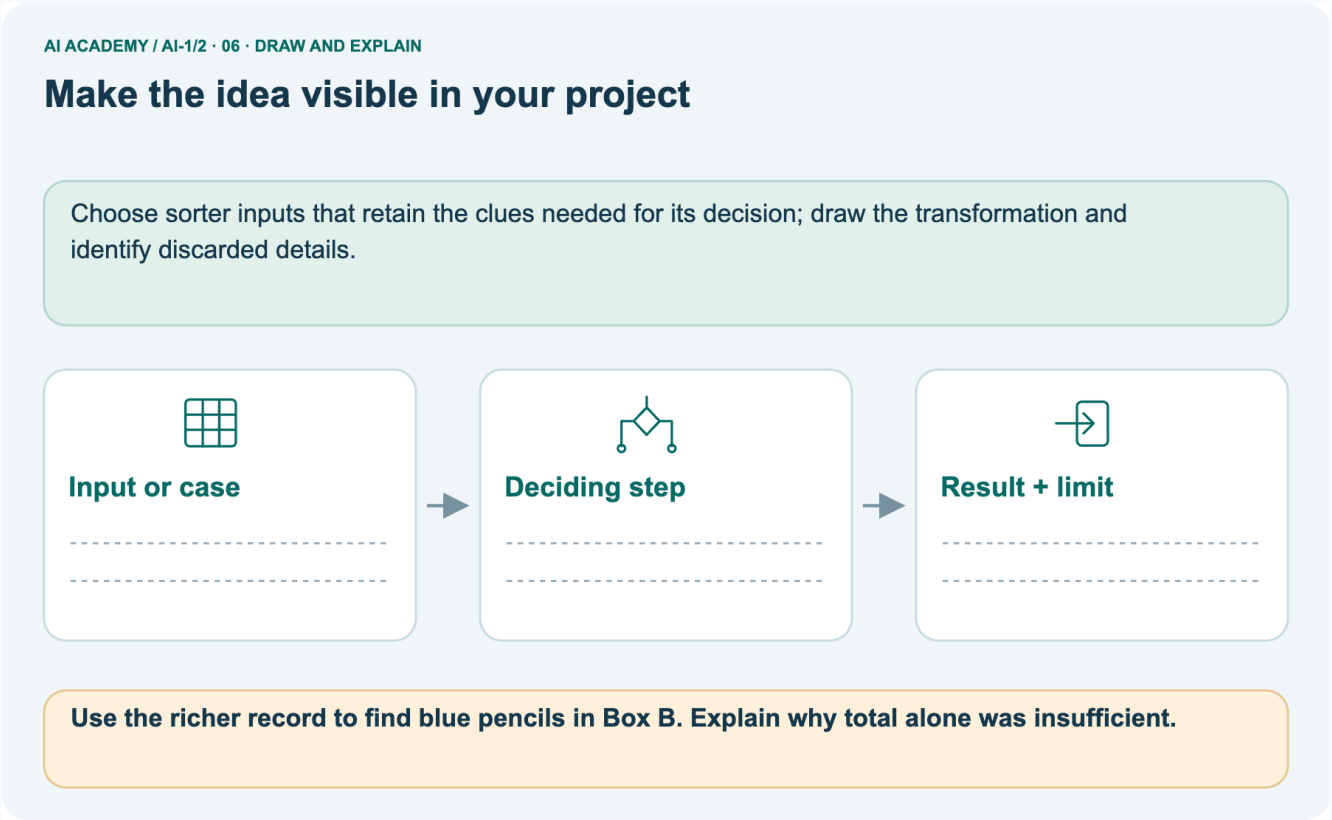
Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct **input**s lead to the same stored **output**.

Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally. Limit: The two-colour assumption matters. Extra colours would prevent inferring blue from total and red alone.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose sorter **input**s that retain the clues needed for its decision; draw the transformation and identify discarded details. Draw your **input**, deciding step, result and limit.

Supported question: Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

Draw your project **trace** and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternative practice and picture transcripts

01. Input, Process, Output: the case

02. Follow the deciding steps

03. Different pencil colours. The same total.

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.